BEFORE YOU READ

Fifteen derivatives in three families

The special derivatives come from three families: the inverse trigonometric, the hyperbolic and the inverse hyperbolic functions. Each entry shows the derivative and where it holds.

L0 **Special derivatives**

L1 **Inverse trigonometric**

L2	**arcsin x** $\lvert x\rvert < 1$	$1 / \sqrt{(1 - x^2)}$
L2	**arccos x** $\lvert x\rvert < 1$	$-1 / \sqrt{(1 - x^2)}$
L2	**arctan x** all real x	$1 / (1 + x^2)$
L2	**arccot x** all real x	$-1 / (1 + x^2)$
L2	**arcsec x** $\lvert x\rvert > 1$	$1 / (\lvert x\rvert \sqrt{(x^2 - 1)})$
L2	**arccsc x** $\lvert x\rvert > 1$	$-1 / (\lvert x\rvert \sqrt{(x^2 - 1)})$

L1 **Hyperbolic**

L2	**sinh x** all real x	$\cosh x$
L2	**cosh x** all real x	$\sinh x$
L2	**tanh x** all real x	$\operatorname{sech}^2 x$
L2	**coth x** $x \neq 0$	$-\operatorname{csch}^2 x$
L2	**sech x** all real x	$-\operatorname{sech} x \cdot \tanh x$
L2	**csch x** $x \neq 0$	$-\operatorname{csch} x \cdot \coth x$

L1 **Inverse hyperbolic**

L2	**arcsinh x** all real x	$1 / \sqrt{(x^2 + 1)}$
L2	**arccosh x** $x > 1$	$1 / \sqrt{(x^2 - 1)}$
L2	**arctanh x** $\lvert x\rvert < 1$	$1 / (1 - x^2)$

TAKEAWAY

Watch the domain

Inverse functions bring square roots and restricted domains; the hyperbolic derivatives mirror the trigonometric ones with sign changes.